UNRAMIFIED AND LOCALLY QUASI-FINITE DETECTED AT THE IDENTITY SECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 05W2: sources/stacks-chow/tags.txt, source lines 7571–7571. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0ABM: sources/stacks-chow/tags.txt, source lines 13025–13025. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let B be an algebraic space over S. Let G be a group algebraic space over B. Assume $G \to B$ is locally of finite type. Then $G \to B$ is unramified (resp. locally quasi-finite) if and only if $G \to B$ is unramified (resp. quasi-finite) at $e(b)$ for all $b \in |B|$.*

Proof. By Morphisms of Spaces, Lemma 05W2 (resp. Morphisms of Spaces, Lemma 0ABM) there is a maximal open subspace $U \subset G$ such that $U \to B$ is unramified (resp. locally quasi-finite) and formation of U commutes with base change. Thus we reduce to the case where $B = \mathrm{Spec}(k)$ is the spectrum of a field. Let $g \in G(K)$ be a point with values in an extension K/k. Then to check whether or not g is in U, we may base change to K. Hence it suffices to show

$$G \to \mathrm{Spec}(k) \text{ is unramified at } e \Leftrightarrow G \to \mathrm{Spec}(k) \text{ is unramified at } g$$

for a k-rational point g (resp. similarly for quasi-finite at g and e). Since translation by g is an automorphism of G over k this is clear. □

Date: stacks-project:spaces-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.